

Telco Customer Churn Analysis Report

1. Introduction

This report presents a detailed analysis of customer churn using the Telco Customer dataset consisting of 7043 customers and 21 features. The purpose of this analysis is to identify the major factors responsible for customer churn and provide data-driven insights that can help improve customer retention strategies.

2. Objective

The main objective of this project is to analyze customer behavior, identify patterns related to churn, and suggest actionable business recommendations to reduce churn rate and improve customer satisfaction.

3. Key Insights

Overall Churn Rate: Approximately 26–27% of customers have churned, while around 73–74% customers are retained. This indicates that more than one-fourth of customers are leaving the service.

Contract Type: Customers with month-to-month contracts have the highest churn rate (~42–45%), while one-year (~11–12%) and two-year (~2–3%) contracts show much lower churn rates.

Payment Method: Customers using electronic check show high churn (~40–45%) compared to credit card or bank transfer users (~15–20%).

Tenure: New customers (0–12 months) have a churn rate above 40%, while long-term customers show significantly lower churn.

Internet Service: Fiber optic users show higher churn (~40%) compared to DSL users (~20%).

Additional Services: Customers without services like online security and tech support have higher churn (~40%) compared to those who use them (~15–20%).

Monthly Charges: Customers with higher monthly charges (>70) are more likely to churn (~35–40%).

Senior Citizens: Senior citizens show higher churn (~40%) compared to non-senior customers (~25%).

Paperless Billing: Customers using paperless billing have higher churn (~33%) than those who do not (~16%).

4. Conclusion

The analysis clearly shows that customer churn is influenced by multiple factors such as contract type, tenure, payment method, service usage, and monthly charges. Customers who are new, paying higher charges, and not using additional services are more likely to churn.

5. Recommendations

To reduce churn, the company should encourage long-term contracts, focus on retaining new customers, promote value-added services, improve service quality for fiber optic users, and provide flexible pricing strategies for high-paying customers.